

Quality of Life Among Parents of Children and youth with Autism Spectrum Disorder: A Cross-Cultural Comparison Between Algeria and France

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Received: 2025-05-20.

Accepted: 2025-06-30.

Published: 2025-07-30

Abstract:

This study aimed to assess and compare the quality of life (QoL) of parents raising children with autism spectrum disorder (ASD) in Algeria and France. Using culturally adapted versions of the Family Quality of Life Scale, the study included two purposive samples of 40 parents from each country. Results revealed a relatively high perceived QoL among parents in both contexts, with no statistically significant difference between the two groups. In Algeria, QoL was rated as upper-moderate, potentially supported by strong informal networks, active involvement in parent associations, cultural and religious coping mechanisms, and recent national autism initiatives. In France, similarly favorable QoL levels were linked to well-established institutional support systems, professional services, and inclusive policies. These findings underscore the multidimensional and culturally sensitive nature of parental well-being and highlight the critical role of both formal and informal support structures. The study calls for comprehensive, culturally responsive policies and programs that reinforce caregiver support and community resilience to enhance outcomes for families of children with ASD.

Keywords: Algeria; autism spectrum disorder; Comparative Study; Cultural Context ; Family Quality of Life; France.

الملخص:

هدفت هذه الدراسة إلى تقييم نوعية الحياة الأسرية لدى أولياء الأطفال المصابين باضطراب طيف التوحد (ASD)، وإجراء مقارنة بين عيّنتين من الجزائر وفرنسا. ولتحقيق ذلك تم الاعتماد على المنهج الوصفي المقارن، واستخدام نسختين من مقياس نوعية الحياة الأسرية (FQoL Scale) بعد تكييفهما وفقًا للسياقين الثقافيين الجزائري والفرنسي. تألفت عينة الدراسة من 80 مشاركًا، الذي تم اختيارهم بالطريقة القصدية، موزعين بالتساوي بين البلدين (40 وليًا من الجزائر و40 وليًا من فرنسا).

وقد أسفرت نتائج الدراسة عن ما يلي:

- أظهرت نتائج العينة الجزائرية مستوى نوعية حياة يتراوح بين المتوسط والمرتفع، وهو ما جاء مخالفًا نسبيًا للفرضية الأولية التي توقعت مستوى متوسطًا. ويرجع هذا المستوى المرتفع نسبيًا لدى الأولياء الجزائريين إلى عوامل مختلفة، أبرزها وجود شبكات دعم اجتماعي وعائلي قوية، والانخراط الفعال لغالبية الأولياء في جمعيات متخصصة بالتوحد، بالإضافة إلى تأثير القيم الثقافية والدينية وأساليب التكيف المجتمعية. كما يُحتمل أن يكون لمبادرات خطة التوحد الوطنية وجهود التوعية المجتمعية الأخيرة دور إيجابي في تعزيز نوعية حياة هذه الأسر.
 - فيما يتعلق بالعينة الفرنسية، فقد أظهرت هي الأخرى مستوى نوعية حياة مرتفعًا بشكل عام، ويفسر ذلك أساسًا بتوفر الدعم المؤسسي الاحترافي، والخدمات المتخصصة، والسياسات الاجتماعية الداعمة والمستقرة التي تعزز مرونة الأسر وتكيفها مع احتياجات أطفالها ذوي التوحد.
 - وعلى الرغم من الاختلافات المتوقعة بين البلدين، لم تظهر فروق ذات دلالة إحصائية في مستوى نوعية الحياة بين العيّنتين، مما يدل على أهمية عوامل الحماية الشخصية والثقافية والاجتماعية بغض النظر عن اختلاف البيئة والسياسات.
- بناءً على هذه النتائج، توصي الدراسة بتعزيز السياسات والبرامج الداعمة بشكل متكامل، بحيث تشمل الجوانب الرسمية وغير الرسمية، مع الأخذ بالاعتبار خصوصية السياق الثقافي والاجتماعي، بهدف تحسين نوعية الحياة على المدى الطويل لأسر الأطفال المصابين باضطراب طيف التوحد

الكلمات المفتاحية: الجزائر - اضطراب طيف التوحد - السياق الثقافي - دراسة مقارنة - فرنسا - نوعية الحياة الأسرية

INTRODUCTION

Autism Spectrum Disorder (ASD) is a complex, lifelong neurodevelopmental condition that significantly affects communication, social interaction, and behavior (American Psychiatric Association, 2013). With a global increase in prevalence, ASD has become a major public health concern, not only because of its impact on individuals but also due to the profound effects it has on family systems, particularly parents (Estes et al. , 2013). This study aims to explore the quality of life (QoL) among parents of children with autism, with a particular focus on comparing French and Algerian families.

The choice of this subject stems from the increasing recognition that parental well-being plays a crucial role in the therapeutic outcomes of children with ASD (Vasilopoulou & Nisbet, 2016). Moreover, while substantial research exists in Western contexts, there remains a lack of comparative cross-cultural studies that investigate how systemic, cultural, and societal factors influence parental experiences. Understanding these differences is essential for developing more effective, culturally sensitive support mechanisms for families.

Research has consistently shown that parenting a child with autism can lead to elevated levels of stress, depression, financial burden, and social isolation (Dardas & Ahmad, 2014; Hayes & Watson, 2013). However, the degree to which these challenges affect parents varies widely across different countries depending on the availability of resources, public policies, and societal attitudes. In France, for example, there is structured support for children with disabilities, although systemic gaps remain (Tchimino, 2019). In Algeria, by contrast, families often confront significant obstacles, including limited access to specialized care and persistent social stigma (Belkacem et al. , 2021).

2- Problem Statement:

Autism Spectrum Disorder (ASD) presents a wide range of lifelong challenges that deeply affect not only individuals but also their families, particularly the parents who often assume primary caregiving responsibilities (Hayes & Watson, 2013). Research has consistently demonstrated that parents of children with autism experience significantly higher levels of stress, emotional distress, and financial strain compared to parents of neurotypical children or those with other developmental disabilities (Karst & Van Hecke, 2012; Davis & Carter, 2008).

However, the degree to which these challenges affect parental quality of life is not uniform across cultural, social, and healthcare contexts. In France, numerous public initiatives — particularly the national "Plans Autisme" — have progressively

strengthened early diagnosis services, therapeutic interventions, inclusive education, and societal awareness about autism (Tchimino, 2019; Pinel-Jacquemin & Gaudron, 2021). Although challenges such as service accessibility and regional disparities persist, structured support mechanisms have contributed to improvements in family experiences.

In Algeria, families historically faced significant difficulties in accessing specialized services and dealing with entrenched societal stigma (Belkacem et al. , 2021; Daoudi et al. , 2020). Nevertheless, recent governmental efforts, marked by the launch of the first National Autism Plan (2021–2024) and preparations for a second plan (2025–2029), reflect a growing commitment to improving the quality of life of individuals with autism and their families. Despite these promising developments, service provision remains uneven, and societal attitudes continue to evolve gradually, particularly outside major urban centers.

Given these contrasting socio-political and healthcare landscapes, several crucial research questions emerge:

2.1- Research questions:

Main research question:

- Is there a significant difference in the level of quality of life experienced by parents of children and youth with autism spectrum disorder between Algerian and French families?

Sub research questions:

- What is the level of quality of life among parents of children and youth with autism spectrum disorder in Algeria?
- What is the level of quality of life among parents of children and youth with autism spectrum disorder in France?

2. 2- Research Hypotheses:

Main Hypothesis:

- There is a statistically significant difference in the level of quality of life between Algerian and French parents of children and youth with autism spectrum disorder.

Sub-Hypothesis:

- The overall quality of life among Algerian parents of children and youth with autism spectrum disorder is at a moderate level.
- The quality of life among French parents of children of youth with autism spectrum disorder is at a high level.

3-Importance of the study:

This research is significant for several reasons. First, it highlights the often-overlooked experiences of parents, who are critical stakeholders in the care and development of children with autism. Second, it offers valuable insights for policymakers, healthcare providers, and advocacy groups aiming to improve support systems for families. Lastly, it contributes to the limited body of comparative, cross-cultural research on autism and family well-being.

4-Objectives of the study:

This study aims to explore the quality of life of parents of children with autism within two distinct socio-cultural and healthcare contexts: Algeria and France. It seeks to better understand how national systems, cultural perceptions, and support infrastructures influence the everyday experiences and well-being of these parents. Specifically, the study intends to evaluate the overall level of quality of life among parents of children with autism in Algeria and France, considering emotional, financial, and social dimensions. Finally, the research seeks to compare the experiences of parents in both countries to determine whether a statistically significant difference exists in their reported levels of quality of life.

5-Concepts of the Study:**5. 1-Autism Spectrum Disorder (ASD):**

Autism Spectrum Disorder (ASD) is a complex neurodevelopmental condition characterized by persistent difficulties in communication and social interaction, along with restricted, repetitive patterns of behavior, interests, or activities (American Psychiatric Association, 2013). The clinical presentation of ASD is highly heterogeneous, with symptoms ranging from mild to severe, affecting cognitive functioning, language development, sensory processing, and daily living skills to

varying degrees.

Globally, the prevalence of autism has seen a steady increase over recent decades. According to the World Health Organization (WHO, 2022), it is estimated that approximately one in every 100 children worldwide is diagnosed with ASD. In the United States, where large-scale surveillance systems such as the Autism and Developmental Disabilities Monitoring (ADDM) Network operate, the Centers for Disease Control and Prevention (CDC) reported a prevalence of approximately 1 in 36 children in 2023 (CDC, 2023). These figures highlight not only growing awareness and better diagnostic practices but also the urgent need for systemic support structures to meet the needs of individuals with ASD and their families.

The impact of ASD on family life is profound. Raising a child with autism often demands continuous attention, intensive behavioral management, and frequent interaction with educational and healthcare institutions. These responsibilities place significant psychological, emotional, and financial burdens on parents (Estes et al., 2013). Many parents report elevated levels of stress, anxiety, depression, and social isolation compared to parents of neurotypical children or those with other disabilities (Karst & Van Hecke, 2012). Furthermore, the unpredictability and long-term nature of autism-related challenges require families to adapt constantly, often necessitating lifestyle changes, career sacrifices, and an ongoing search for appropriate interventions.

In this context, the concept of parental quality of life becomes central. It encompasses how parents perceive and experience their own well-being while managing the complex realities of raising a child with autism. Understanding the variability in parental experiences—across different cultural, economic, and healthcare systems—is critical for developing supportive policies and interventions that not only address the needs of children with ASD but also promote family resilience and well-being.

5. 1. 1-Autism in Algeria: Current Situation

In Algeria, autism spectrum disorder (ASD) is an increasingly recognized neurodevelopmental condition. According to the Algerian Ministry of Health, approximately 450, 000 people are currently living with autism across the country, suggesting a national prevalence that exceeds global averages and reflects either more inclusive diagnostic criteria or late identification (Ministère de la Santé, 2023). Historically, autism remained largely misunderstood and underdiagnosed in Algeria.

Families, especially in rural regions, often faced the journey of diagnosis and support with little to no institutional guidance.

Until recently, the country lacked a coherent national framework for autism care. Services were fragmented, concentrated in major urban centers, and heavily reliant on out-of-pocket expenses. Qualified professionals—particularly child psychiatrists, developmental psychologists, and specialized educators—remain insufficient in number. As a result, many children are diagnosed late, and families experience considerable emotional and financial stress.

However, significant progress has begun. The First National Autism Plan (2021–2024) marked a turning point by prioritizing early detection, professional training, and public awareness. The government is currently preparing a Second National Plan for Autism (2025–2029), aiming to scale up services, standardize care protocols, and strengthen intersectoral coordination between health, education, and social affairs. Despite these efforts, serious challenges remain. The implementation of autism services is uneven across regions, particularly outside the major cities, and stigma rooted in cultural misconceptions continues to limit social inclusion and early intervention.

5. 1. 2-Autism in France: Current Situation

France reports an estimated autism prevalence of 1 in 100 children, in line with international statistics (Santé Publique France, 2020). Over the past two decades, France has significantly expanded its national autism policy through four successive “Plans Autisme,” starting in 2005. These state-led initiatives have aimed to improve early screening, inclusive education, multidisciplinary care, and the support of families. The most recent Plan Autisme 2018–2022, framed under the broader “Strategy for Neurodevelopmental Disorders,” placed particular emphasis on diagnosis before the age of three, integration into school environments, and professional development.

Families in France generally benefit from institutional support, such as early medical screening, disability allowances (*AEEH*), and publicly funded services including speech therapy, occupational therapy, and behavioral interventions. Additionally, parent associations and NGOs play an important role in advocacy and service provision. Public health insurance systems reduce out-of-pocket costs for families, though private therapies are often necessary to compensate for delays in public care.

Nonetheless, disparities remain. Service accessibility varies considerably by

region, and rural areas still suffer from long waiting times and a shortage of qualified professionals. Furthermore, while social attitudes toward autism have improved over the years, some families report ongoing administrative difficulties, a lack of individualized care pathways, and persistent misconceptions within the education system. Overall, France is recognized as a relatively advanced country in terms of autism policy, though it continues to face challenges in achieving equitable and timely access to care.

5. 2-Quality of Life (QoL):

The central concept of this study is quality of life (QoL), a multidimensional construct that encompasses an individual's subjective evaluation of their physical health, emotional well-being, social relationships, level of independence, financial security, and interaction with their environment. According to the World Health Organization (1997), QoL is defined as *“an individual's perception of their position in life in the context of the culture and value systems in which they live, and in relation to their goals, expectations, standards and concerns.”*

In the context of autism, QoL becomes particularly relevant for parents who assume long-term caregiving responsibilities. Raising a child with autism spectrum disorder (ASD) often involves navigating complex healthcare systems, educational challenges, social stigma, and financial burdens, all of which may significantly impact parental well-being (Vasilopoulou & Nisbet, 2016). Parents of children with autism tend to report lower QoL scores than the general population, especially in psychological and social domains (Derguy et al. , 2015).

Moreover, the experience of QoL is shaped by the availability and accessibility of services, the presence of supportive networks, and the degree of societal inclusion. In environments where autism is poorly understood or stigmatized, parental QoL is further diminished due to feelings of isolation and lack of support (Gray, 2002; Dardas & Ahmad, 2014). As such, QoL is not only a reflection of personal resilience but also of external factors that either support or hinder the caregiving journey.

5. 3- Family Quality of Life (FQOL):

While QoL often focuses on the individual, the concept of Family Quality of Life (FQOL) provides a broader framework that examines the collective well-being of all family members living with a child with a disability. FQOL is defined as *“a dynamic sense of well-being of the family, collectively and subjectively defined and informed by its members, in which individual and family-level needs interact”* (Zuna et al. , 2010).

For families of children with autism, FQOL includes factors such as family cohesion, emotional support, adaptability, satisfaction with caregiving roles, access to services, and the ability to participate in community life (Brown et al. , 2006). Research shows that families of children with ASD often report lower FQOL than those raising children with other disabilities, particularly due to higher caregiving

demands, behavioral challenges, and long-term uncertainty (Hu et al. , 2012).

The FQOL framework is especially relevant for cross-cultural studies, as family dynamics, expectations, and support systems vary significantly across societies. In cultures with strong extended family networks and shared caregiving responsibilities, the burden may be more distributed, whereas in societies with limited social infrastructure, families may experience more strain. Therefore, measuring FQOL in both Algerian and French contexts allows for a more nuanced understanding of how cultural and systemic factors shape the everyday lives of families raising children with autism.

6- Methodological Procedures of Field Study:

6. 1-Study Methodology:

This study employs a descriptive and comparative methodology to assess and compare the quality of life of parents raising children and youth with autism spectrum disorder (ASD) in Algeria and France. The descriptive component enables a thorough analysis of their living conditions, while the comparative approach highlights the similarities and differences between the two national contexts.

6. 2- Study Boundaries:

6.2.1- Temporal Boundaries: The study was conducted between September 2023 and June 2024.

6.2.2- Spatial Boundaries: The study was carried out in several Algerian and French cities.

Graph (01): Residence distribution in the Algerian sample

The bar chart illustrates the distribution of participants' residence locations within the Algerian sample. The data reveal a strong concentration in urban and more populous areas. Algiers accounts for the highest proportion of respondents, representing over 35% of the sample, followed by Boumerdes at approximately 22%.

Other regions such as Mila, Chlef, and Bejaia contribute modestly to the sample, each representing between 5% and 10%.

In contrast, several other cities—including Tizi Ouzou, Oran, Blida, Constantine, and Souk Ahras—show minimal representation, each contributing less than 5% of the total. This skewed distribution suggests that the sample is predominantly drawn from the central northern region of Algeria, particularly the capital and surrounding wilayas, which may reflect better access to autism-related services or higher awareness in these areas.

Graph (02): Residence distribution in the French sample

The bar chart displays the geographic distribution of participants in the French sample. The data indicates a notable concentration in the Val d'Oise, which alone accounts for nearly 25% of the total respondents. This high representation suggests a possible clustering of participants in the Île-de-France region, likely due to greater access to diagnostic and support services, or the presence of active autism-related networks in this area.

Other departments such as Ain, Hauts-de-Seine, Seine-Saint-Denis, and Meurthe-et-Moselle show moderate representation, each contributing between 4% and 7% of the sample. In contrast, the majority of other regions—including Gironde, Côtes-d'Armor, Isère, and Landes—are represented at low levels, below 5%.

This uneven residential distribution highlights a centralization of participants in urban and suburban areas of northern France, particularly around Paris.

6. 3- Study Population and Sample: The study included a total of 80 parents of children and youth diagnosed with autism spectrum disorder (ASD), with 40 participants from Algeria and 40 from France. A purposive sampling approach was used to ensure relevance and diversity within the sample, which included Algerian and French families with a child or a youth diagnosed with ASD.

– Sample characteristics:

Table (01): Parent's characteristics

Parent Characteristic	Algerian Sample	French Sample
Most Common Fathers' Age Group	45–55 years (40%)	35–45 years (52. 5%)
Most Common Mothers' Age Group	35–45 years (65. 5%)	35–45 years (55%)
Fathers with University Education	52. 5%	50%

Mothers with University Education	45%	65%
Marital Status (Married)	85%	70%
Most Common Number of Children	4 children (35%)	2 children (45%)
Living Standard (Medium)	80%	50%

The table above presents the key sociodemographic characteristics of parents in the Algerian and French samples. In terms of age, Algerian fathers were mostly aged 45–55 years (40%), whereas 52. 5% of French fathers were in the younger 35–45-year group. A similar pattern was seen among mothers, with 65. 5% of Algerian mothers aged 35–45, compared to 55% in France, suggesting a relatively older parenting profile in Algeria.

Educational attainment also showed variation between the two samples. In Algeria, 52. 5% of fathers and 45% of mothers held university degrees, compared to 50% of French fathers and a significantly higher 65% of French mothers. These differences may affect parents' ability to access and navigate specialized services.

In terms of marital status, the majority of Algerian parents were married (85%), compared to 70% in France, where the proportion of divorced individuals was notably higher.

Family size further distinguished the two groups: 35% of Algerian families had four children, while 45% of French families had two, indicating a generally larger household size in the Algerian context.

Finally, regarding living standards, 80% of Algerian parents reported a medium level, whereas in France, only 50% reported the same, with 37. 5% indicating a good living standard. These socioeconomic differences are important factors to consider when interpreting quality of life outcomes.

Table (02): Child's characteristics

Child Characteristic	Algerian Sample	French Sample
Most Common Age Group	10–15 years (60%)	5–10 years (62. 5%)
Gender Distribution (Male)	Above 80%	85%
Most Common Age at Diagnosis	2–5 years (close to 100%)	2–5 years (62. 5%)
Access to Specialized Care	90% have access	97. 5% have access

The table above summarizes the main characteristics of children with autism in

both the Algerian and French samples. Most Algerian children were aged 10–15 years (60%), while in France, the majority fell within the 5–10-year age range (62.5%). Gender distribution was similar across both groups, with more than 80% of children being male, reflecting the global trend of higher ASD prevalence among boys.

Interestingly, age at diagnosis appeared earlier in Algeria, where nearly all children (approximately 100%) were diagnosed between the ages of 2 and 5, compared to 62.5% in the French sample. Lastly, access to specialized care was reported by 90% of Algerian parents and 97.5% of French parents, indicating broader coverage in France but also notable progress in service availability within the Algerian context.

6. 4- Data Collection Tools:

The instrument used for data collection in this study was the Family Quality of Life Scale (FQoL), originally developed by the Beach Center on Disability. This scale evaluates multiple dimensions of family well-being, including emotional support, financial stability, family interaction, physical/material well-being, and access to services.

Two culturally appropriate versions of the scale were employed. For the Algerian sample, a version of the FQoL scale was adapted and standardized by the researcher, taking into account linguistic and cultural nuances specific to the Algerian context. For the French sample, the study used the validated French version of the scale developed by Mélina Rivard, which has been widely applied in Francophone research settings involving families of children with developmental disabilities.

The use of two context-specific versions of the FQoL scale ensured that the data collected reflected the lived realities of families in both Algeria and France, while maintaining consistency in the core dimensions assessed.

6. 4. 1-Reliability and Validity of the Scales: The Family Quality of Life Scale (FQoL) used in this study demonstrated strong psychometric properties in both versions.

For the Algerian version, which was linguistically and culturally adapted by the researcher, reliability was assessed using Cronbach's Alpha, yielding an excellent internal consistency score of 0.941 across the 25 items. Construct validity was confirmed through Pearson correlation coefficients between items and their respective subdimensions, all of which were statistically significant at the 0.01 level. These results confirm that the Algerian version of the scale is both reliable and valid, and suitable for assessing family quality of life within the Algerian context.

Table (03): Psychometric properties of the Algerian version of the FqoL Scale

Psychometric Property	Algerian Version
Sample Size	70 Algerian parents of children with ASD
Factor Structure	Five factors: Family Interaction, Parenting, Emotional Well-being, Material Well-being, Disability-related Support
Internal Consistency (Cronbach's Alpha)	0. 941 (Total scale)
Construct Validity (Pearson Correlations)	All items significantly correlated with their subdimensions (0. 58 -0. 89)
Significance Level	$p < 0. 01$

The French version of the Family Quality of Life Scale (FQoL), developed and validated by Rivard et al. (2017), was tested on a large sample of 452 French-speaking parents of children recently diagnosed with autism spectrum disorder (ASD). The study confirmed strong psychometric properties across several domains.

A confirmatory factor analysis (CFA) supported the original five-factor structure of the scale—family interaction, parenting, emotional well-being, material well-being, and disability-related support. Goodness-of-fit indices indicated acceptable model fit (e. g. , CFI = 0. 93, RMSEA = 0. 06), confirming the structural validity of the scale.

The internal consistency of the Satisfaction and Importance dimensions was found to be excellent. The Cronbach's alpha coefficients for the global Satisfaction scale reached 0. 92, and for the Importance scale 0. 90, indicating high reliability. Subscale alphas ranged from 0. 71 to 0. 88, reflecting acceptable to excellent reliability across domains.

Table (04): Psychometric properties of the French version of FqoL Scale

Psychometric Property	French Version
Sample Size	452 French-speaking parents of children with ASD
Factor Structure	Five factors: Family Interaction, Parenting, Emotional Well-being, Material Well-being, Disability-related Support

Goodness-of-Fit Indices	CFI = 0. 93, RMSEA = 0. 06
Global Satisfaction Reliability (Cronbach's Alpha)	0. 92
Importance Scale Reliability (Cronbach's Alpha)	0. 90
Subscale Reliability (Cronbach's Alpha Range)	0. 71 – 0. 88

7- Presentation and Interpretation of Study Results:

7. 1. Presentation and interpretation of the First Sub-Hypothesis Results:

- The overall quality of life among Algerian parents of children and youth with autism spectrum disorder is moderate.

To verify the validity of this hypothesis, percentages, means, and standard deviations were calculated for the responses of the sample participants to the items of the Family Quality of Life Scale.

Graph (03): Quality of life level among parents of the Algerian sample

The graph above presents the results of the Family Quality of Life Scale (FQoL) as reported by Algerian parents of children with autism spectrum disorder (ASD). The data suggest that the overall quality of life among these parents is predominantly high to moderate. Specifically, 55% of respondents reported a high level of quality of life, 40% indicated a moderate level, and only 5% reported a low level of well-being. These findings reflect a relatively positive perception of family life despite the challenges associated with caregiving.

Table (05): Descriptive Statistics of Family Quality of Life (FQoL) Scores

(Algerian sample)

Variable	N	Minimum	Maximum	Mean	Standard Deviation
Total FQoL	40	32.00	113.00	88. 60	17.89
Valid N	40				

The table below shows that the mean Family Quality of Life (FQoL) score of 88.6 out of 125 situates the Algerian sample within the upper-moderate range of perceived quality of life. Scores varied considerably, ranging from a minimum of 32 to a maximum of 113, highlighting significant individual differences. However, the moderate standard deviation (17.89) indicates that perceptions of quality of life were relatively consistent across the majority of respondents.

These findings challenge the initial hypothesis, which predicted that Algerian parents would predominantly report moderate or lower quality of life due to structural and economic constraints. Instead, the results suggest higher-than-anticipated levels of parental well-being. This positive outcome may be influenced by contextual and cultural factors, including informal support networks, community solidarity, religious and spiritual coping mechanisms, and adaptive family dynamics. Additionally, it is important to note that the majority of the Algerian sample consisted of parents affiliated with parent associations, likely enhancing their emotional, informational, and advocacy resources. Moreover, Algeria's recent implementation of the National Autism Plan (2016–2020) and increased public awareness initiatives may also have fostered a more supportive and understanding societal context, thereby positively influencing parents' experiences.

These findings are consistent with previous studies, such as those by Wang et al. (2018) and Vasilopoulou and Nisbet (2016), who found that despite significant emotional, financial, and social burdens, many parents of children with autism maintain relatively high levels of quality of life, primarily due to protective factors

like family cohesion, access to specialized services, and psychosocial support systems.

However, despite these cultural and informal support mechanisms, structural gaps remain evident. As highlighted by Sahraoui et al. (2022), families in Algeria continue to face considerable challenges, including limited specialized healthcare services, financial pressures, and insufficient societal awareness of autism. Such factors likely continue to affect vulnerable subgroups disproportionately.

Ultimately, the divergence between the initial hypothesis and actual results

underscores the complex and culturally nuanced nature of family quality of life. While protective factors clearly mitigate potential declines in quality of life, the fact that 40% of parents still reported moderate levels of quality of life, and 5% reported low levels, emphasizes the continued need for targeted interventions. Future efforts should prioritize improving healthcare infrastructure, increasing financial assistance, and developing inclusive policies and caregiver-centered programs that holistically address family needs and promote sustained resilience and well-being.

7. 2- Presentation and interpretation of the Second Sub-Hypothesis Results:

- The quality of life among French parents of children of youth with autism spectrum disorder is relatively high.

Graph (04): Quality of life level among parents of the French sample

The corresponding graph for the French sample reveals a similar distribution pattern, with slightly higher indicators of well-being. According to the results, 57. 5% of French parents reported a high level of quality of life, 37. 5% indicated a moderate level, and 5% reported a low level. These outcomes suggest that the majority of French parents of children with ASD experience a favorable quality of life, potentially supported by broader access to services and resources.

Table (06): Descriptive Statistics of Family Quality of Life (FQoL) Scores (French sample)

Variable	N	Minimum	Maximum	Mean	Standard Deviation
Total FQoL	40	51. 00	125. 00	92. 50	16. 08
Valid N	40				

The table below shows that the mean Family Quality of Life (FQoL) score of 92. 5 out of 125 places the French sample within the high range of perceived quality of life. The scores range from a minimum of 51 to a maximum of 125, reflecting a broad spectrum of parental experiences. The standard deviation of 16. 08 indicates moderate variability, suggesting that while many participants reported high levels of well-being, there are notable individual differences in their lived realities.

These results align with expectations, as previous research has shown that parents in countries with well-established social and healthcare systems—such as France—tend to report higher quality of life. Contributing factors may include greater access

to specialized services, financial support programs, professional interventions, and a more inclusive educational environment. The high level of parental education, particularly among mothers, likely plays a role in facilitating more effective coping strategies and greater engagement with available resources (Pisula & Porębowicz-Dörsmann, 2017).

Supporting this interpretation, Cappe et al. (2020) found that access to governmental and professional support services is a critical determinant of parental well-being among families of children with ASD in France. Likewise, Eapen et al. (2014) emphasized the impact of comprehensive service infrastructures, caregiver education, and emotional support in improving family outcomes.

The data further reveal that 37.5% of parents fall within the moderate QoL range, and 5% within the low range, suggesting that disparities still exist, despite the generally favorable context. These gaps may be due to regional differences in service provision, financial stress, or variations in the severity of the child's condition. As Bourke-Taylor et al. (2014) noted, even in resource-rich settings, caregiver burden and unmet needs persist, especially among families with limited social or familial support.

In conclusion, while the overall quality of life among French parents appears high, the variability in scores highlights the need to expand equitable access to care, reduce systemic inequalities, and promote inclusive, family-centered services. These findings support a more nuanced understanding of parental well-being and the importance of tailoring interventions to diverse family needs.

7. 3- Presentation and interpretation of the main Hypothesis Results:

- There is a statistically significant difference in the level of quality of life between Algerian and French parents of children and youth with autism spectrum disorder.

To verify the validity of this hypothesis, an independent samples **t-test** was used to compare the mean quality of life scores between the Algerian and French samples.

Table (07): Independent Samples t-Test Results with Levene's Test for Equality of Variances

Hypothesis	Levene's Test F	Sig	t-value	d f	Sig. (2-tailed)	Mean Difference	Std. Error Difference	95% Confidence Interval of the Difference (Lower – Upper)
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Equal variances assumed	0.074	0.786	0.926	78	0.358	6.236	6.741	-7.181 to 19.658
Equal variances not assumed	—	—	0.980	73.85	0.358	6.239	6.365	-8.654 to 21.132

The results of the independent samples t-test presented in Table (07) indicate that there is no statistically significant difference in the mean Family Quality of Life (FQoL) scores between the Algerian and French samples. Levene's test for equality of variances yielded an F-value of 0.074 with a significance level of 0.786, suggesting that the assumption of equal variances is met.

Using the equal variances assumed row, the t-value is 0.926 with 78 degrees of freedom, and the two-tailed significance (p-value) is 0.358, which is well above the conventional threshold of 0.05. The mean difference between the two groups is 6.236, with a standard error of 6.741, and the 95% confidence interval ranges from -7.181 to 19.658. This interval includes zero, further confirming that the difference is not statistically significant.

These findings suggest that, although the French sample had a slightly higher mean QoL score, the difference is not large enough to be considered statistically meaningful. Therefore, the data support the conclusion that parents of children with

ASD in both Algeria and France report comparable levels of family quality of life, despite differences in socioeconomic and healthcare contexts.

The independent samples t-test revealed no statistically significant difference in the mean Family Quality of Life (FQoL) scores between Algerian and French parents of children with autism spectrum disorder (ASD), with a p-value of 0.358 ($t = 0.926$, $df = 78$). Although the French sample had a slightly higher mean (92.5 vs. 88.6), the confidence interval [-7.181, 19.658] includes zero, confirming that this difference is not statistically meaningful.

These findings are consistent with recent literature suggesting that quality of life among parents of children with ASD is shaped more by individual and familial protective factors than by country of residence alone. For instance, Vasilopoulou & Nisbet (2016) emphasized the central role of coping strategies, perceived support, and psychological resilience in maintaining parental well-being, regardless of socioeconomic context. Similarly, Wang et al. (2018) demonstrated that emotional adaptation and family cohesion can significantly mediate the impact of caregiving stress across different cultures.

The lack of a significant difference between the Algerian and French groups may

reflect the compensatory role of informal supports in the Algerian context—such as religious beliefs, extended family involvement, and cultural values promoting collective caregiving—which may offset the relative scarcity of formal services. This aligns with findings from Gray (2002) and Dardas & Ahmad (2014), who showed that cultural resilience and strong kinship structures in non-Western settings often serve as buffers against caregiver stress.

On the French side, the structured healthcare and social support systems, as highlighted by Cappe et al. (2020), contribute to higher QoL outcomes, but do not necessarily produce radically different experiences when compared to families in more resource-constrained environments. This reinforces the view that QoL is multidimensional and influenced by psychosocial, not only material, factors (Zuna et al. , 2010).

In light of these results, it appears that context-specific strengths in each country may be contributing in different ways to sustaining family well-being. While France offers institutional support, Algeria may benefit from community-based resilience mechanisms. The implication is that interventions aimed at enhancing family QoL should not solely focus on structural reforms but also consider cultural strengths, family dynamics, and emotional resources.

8- Conclusion:

This comparative study examined the quality of life of parents raising children with autism spectrum disorder (ASD) in Algeria and France, emphasizing the complex interplay between cultural, social, and institutional factors. Contrary to initial expectations, no substantial differences emerged between the two groups, as both reported generally favorable perceptions of family well-being.

These outcomes suggest that quality of life is not solely determined by the availability of formal services or material resources. Instead, it appears significantly influenced by cultural values, community support systems, and individual coping strategies. In Algeria, most parents belonged to active parent associations, which likely provided essential emotional, informational, and practical support, helping them cope effectively with caregiving challenges. Additionally, recent initiatives such as Algeria's National Autism Plan and increased public awareness may have contributed positively by fostering a more understanding and inclusive societal environment.

In France, structured institutional support—including professional services, financial assistance, and inclusive educational policies—continues to play a vital role in maintaining parental resilience and overall family well-being. Nevertheless, the similarity in quality of life between the two samples highlights the equally important role of informal supports and culturally embedded resilience mechanisms.

Ultimately, these findings underline the multidimensional and culturally sensitive nature of parental quality of life, advocating for policies and interventions that not only enhance formal services but also nurture and leverage community solidarity and informal support networks. Future research and practice should continue to prioritize culturally adapted, inclusive, and family-centered strategies, recognizing the diverse pathways through which families sustain their resilience and overall well-being.

9-Recommendations and limits of the study:

9. 1- Recommendations:

- Promote family-centered approaches: Interventions should address the well-being of the entire family, not only the child. This includes psychological support, caregiver training, and access to respite services.
- Integrate cultural and contextual sensitivity in service delivery: Support mechanisms should account for local beliefs, informal caregiving traditions, and variations in access to services. In Algeria, for instance, community resilience and extended family involvement may serve as valuable assets.
- Support parent associations: These organizations offer emotional, informational, and advocacy resources. Their role should be institutionally recognized and financially supported in both contexts.
- Reduce geographic inequalities: In both Algeria and France, access to services is unevenly distributed. Expanding service availability in underserved areas, particularly rural regions, is essential.
- Encourage longitudinal and mixed-methods research: Future studies should explore how parental quality of life evolves over time and investigate the underlying narratives through both quantitative and qualitative methods.

9. 2- Limitations

- Sample size and representativeness: With 80 participants selected through purposive sampling, generalizability is limited. The findings should be viewed as exploratory rather than conclusive.

- Urban sampling bias: Both samples predominantly reflect urban and peri-urban populations, potentially overrepresenting families with greater awareness or access to autism services.
- Self-reported measures: All data were collected through self-administered questionnaires, which may be affected by recall or social desirability bias, especially in cultural settings where expressing distress is stigmatized.
- Cross-sectional design: The study offers a snapshot rather than a developmental perspective. It does not capture changes in family experiences over time.

Despite these limitations, the study offers valuable insight into the role of cultural and systemic factors in shaping the quality of life of parents raising children with ASD. It underscores the need for policies that are not only structurally robust but also culturally informed and family-centered.

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